

Question # 1 of 10 (Start time: 01:31:34 PM, 04 July 2023)

Total Marks: 1

Which is the R command for selecting a sample of size 3 from the population `yp <- c(11,150, 121, 198, 12, 136, 14, 129, 17, 115, 186, 110, 121, 15, 14)`:

Select the correct option

☒ `ys <- sample(yp,3)`☐ `ys <- sample(yp,27)`☐ `ys <- sample(yp,9)`☐ `ys <- sample(yp,12)`

Show Answer

Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 01:34:02 PM, 04 July 2023)

Total Marks: 1

What is the bias of ratio estimator in two stage sampling?

Select the correct option

 Reload Math Equations☐

$$MSE(\bar{y}_{r2s}) \approx \bar{Y}^2 (V_{02} - V_{20} - 2V_{11})$$

☐

$$MSE(\bar{y}_{r2s}) \approx \bar{Y}^2 (V_{02} - V_{20} + 2V_{11})$$

☒

$$MSE(\bar{y}_{r2s}) \approx \bar{Y}^2 (V_{02} + V_{20} - 2V_{11})$$

$$MSE(\bar{y}_{r2s}) \approx \bar{Y}^2 (V_{02} + V_{20} - 2V_{11})$$

☐

None of the above

Click to Save Answer & Move to Next Question

Show Answer

Question # 2 of 10 (Start time: 01:32:16 PM, 04 July 2023)

Total Marks: 1

If $N = 4000$, $N_1 = 2000$; $n_1 = 150$, $N_2 = 1500$; $N_3 = 500$; $n_2 = 8$; $n_3 = 25$, the value of W_3 is:

Select the correct option

☐ 0.127☐ 0.50☐ 0.23☒ 0.125

Show Answer

Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 01:34:02 PM, 04 July 2023)

Total Marks: 1

What is the bias of ratio estimator in two stage sampling?

Select the correct option

 Reload Math Equations

- ☐ $MSE(\bar{y}_{r2s}) \approx \bar{Y}^2 (V_{02} - V_{20} - 2V_{11})$
- ☐ $MSE(\bar{y}_{r2s}) \approx \bar{Y}^2 (V_{02} - V_{20} + 2V_{11})$
- ☒ $MSE(\bar{y}_{r2s}) \approx \bar{Y}^2 (V_{02} + V_{20} - 2V_{11})$
- ☐ None of the above

$$MSE(\bar{y}_{r2s}) \approx \bar{Y}^2 (V_{02} + V_{20} - 2V_{11})$$

[Click to Save Answer & Move to Next Question](#)

Show Answer

Question # 5 of 10 (Start time: 01:37:04 PM, 04 July 2023)

Total Marks: 1

If $\bar{y}_1 = 122.8$, $\bar{y}_2 = 134.6$, $\bar{y}_3 = 133.8$ and $\bar{y}_4 = 127.2$, then What will be the means of cluster means for three clusters?

Select the correct option

[Reload Math Equations](#)☐

127.5

☒

128.2

☐

132.5

☐

126.5

Show Answer

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Question # 6 of 10 (Start time: 01:38:34 PM, 04 July 2023)

Total Marks: 1

The below expression for variance of mean estimator under ranked set sampling is:

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---|
| ☐ | $V(\bar{X}^*) = \frac{1}{nk} \sum_{i=1}^k E(X_{(i)}^* - \bar{X}_{(i)}^*)^2 - \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2$ |
| ☒ | $V(\bar{X}^*) = \frac{1}{k^2} \sum_{i=1}^k E(X_{(i)}^* - \bar{X}_{(i)}^*)^2 - \frac{1}{k^2} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2$ |
| ☐ | $V(\bar{X}^*) = \frac{1}{k} \sum_{i=1}^k E(X_{(i)}^* - \bar{X}_{(i)}^*)^2 - \frac{1}{k} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2$ |
| ☐ | $V(\bar{X}^*) = \frac{1}{nk} \sum_{i=1}^k E(X_{(i)}^* - \bar{X}_{(i)}^*)^2 - \frac{1}{m} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2$ |

Show Answer

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 01:40:11 PM, 04 July 2023)

Total Marks: 1

iff:

$$\text{Var}(\bar{y}_{str}) < \text{Var}(\bar{y}_{sy})$$

Select the correct option

 Reload Math Equations

$$\rho_{\text{rest}} > 0$$



$$\rho_{\text{rest}} = 0$$



None of above



$$\rho_{\text{rest}} < 0$$

Show Answer

 Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 01:37:04 PM, 04 July 2023)

Total Marks: 1

If $\bar{y}_1 = 122.8$, $\bar{y}_2 = 134.6$, $\bar{y}_3 = 133.8$ and $\bar{y}_4 = 127.2$, then What will be the means of cluster means for three clusters?

Select the correct option

[Reload Math Equations](#)☐ 127.5☒ 128.2☐ 132.5☐ 126.5

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[Click to Save Answer & Move to Next Question](#)

Question # 8 of 10 (Start time: 01:41:38 PM, 04 July 2023)

Total Marks: 1

$$\text{Var}(\bar{y}_{sy}) < \text{Var}(\bar{y}_{str})$$

Select the correct option

 Reload Math Equations☐

$$\rho_{\text{rest}} > 0$$

☐

None of above

☒

$$\rho_{\text{rest}} < 0$$

☐

$$\rho_{\text{rest}} = 0$$

Click to Save Answer & Move to Next Question

Show Answer

Question # 9 of 10 (Start time: 01:42:04 PM, 04 July 2023)

Total Marks: 1

What is the value of combined type ratio estimator with $\bar{y}_1 = 18.5$, $\bar{y}_2 = 80.25$ and $\bar{y}_3 = 75.32$?

Select the correct option

☐ 15.836☐ 21.521☐ 20.383☒ 17.363

Show Answer

[Click to Save Answer & Move to Next Question](#)

Question # 1 of 10 (Start time: 12:43:56 PM, 21 July 2023)

Total Marks: 1

In sampling without replacement, an element can be chosen:

STA632

Quiz # 4

Select the correct option

 Reload Math Equations

- ☐ Less than once
- ☒ only once
- ☐ More than once
- ☐ Difficult to tell

Show Answer

 Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 12:43:56 PM, 21 July 2023)

Total Marks: 1

In sampling without replacement, an element can be chosen:

STA632

Quiz # 4

Select the correct option

 Reload Math Equations

☐	Less than once
☒	only once
☐	More than once
☐	Difficult to tell

Show Answer

 Click to Save Answer & Move to Next Question

Question # 2 of 10 (Start time: 12:45:11 PM, 21 July 2023)

Total Marks: 1

When the population size is what size, it is recommended to use the entire population instead of a sample?

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|-----------------|
| ☒ | 100 or less |
| ☐ | 1000 or greater |
| ☐ | 100 or greater |
| ☐ | 1000 or less |

Show Answer

 Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 12:46:48 PM, 21 July 2023)

Total Marks: 1

What is the R command to find out variance of x?

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|--------------|
| ☒ | Var (x) |
| ☐ | X = variance |

Show Answer

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 12:47:40 PM, 21 July 2023)

Total Marks: 1

What is the value of combined type ratio estimator with $\bar{y}_u = 15.5$, $\bar{x}_u = 70.25$ and $\bar{x} = 65.32$?

Select the correct option

14.50



13.50

Q. Find the expression of Bias of **Combined Type Ratio Estimator**.

Answer:

➤ **Combined Type Ratio Estimator** in Stratified Sampling

- The **combined type ratio estimator** is

$$\bar{y}_{nr} = \frac{\bar{y}_u}{\bar{x}_u} \bar{x} = 15.5/70.25 * 65.32 = 14.41224$$

Reload Math Equations

Show Answer

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 12:49:24 PM, 21 July 2023)

Problem of non-response

Select the correct option

☐	none	Q . Define Nonresponse? Ans: The individuals chosen for the sample are not ready to participate in the survey. It is called nonresponse. This is a type of selection bias.
☐	no solution	
☐	no meaning	
☒	can be solved	



Click to Save

Question # 5 of 10 (Start time: 12:49:24 PM, 21 July 2023)

Problem of non-response

Select the correct option

☐	none	Q . Define Nonresponse? Ans: The individuals chosen for the sample are not ready to participate in the survey. It is called nonresponse. This is a type of selection bias.
☐	no solution	
☐	no meaning	
☒	can be solved	



Click to Save

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STA632 - Sampling Techniques (Quiz 4)

Quiz Start Time: 12:43 PM, 21 July 2023

Question # 6 of 10 (Start time: 12:50:54 PM, 21 July 2023)

Total Marks: 1

If the respondents do not provide the required information to the researcher, then it is known as

Select the correct option

Reload Math Equations

Q . Define Nonresponse?

Ans: The individuals chosen for the sample are not ready to participate in the survey. It is called nonresponse. This is a type of selection bias.

☒ problem of non-response

☐ none of the above

☐ non-sampling error

☐ both of the above

Show Answer

Click to Save Answer & Move to Next Question

LEVELS UP
WITH
LEVELS ACADEMY
ARIIPWALA

Question # 7 of 10 (Start time: 12:51:56 PM, 21 July 2023)

Total Marks: 1

What is the value of combined type ratio estimator with $\bar{y}_s=18.5$, $\bar{x}_s=80.25$ and $\bar{X}=75.32$?

Select the correct option

[Reload Math Equations](#)

☐	16.323	Q. Find the expression of Bias of Combined Type Ratio Estimator. Answer: ➤ Combined Type Ratio Estimator in Stratified Sampling The combined type ratio estimator is $\bar{y}_{nr} = \frac{\bar{y}_s}{\bar{x}_s} \bar{X}$
☐	15.632	
☒	17.363	
☐	19.364	

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Question # 8 of 10 (Start time: 12:53:29 PM, 21 July 2023)

Total Marks: 1

Who introduced the basic concept of ranked set sampling?

Page#194

Select the correct option

[Reload Math Equations](#)

- ☐ Muhante (1952)
- ☐ Isaki (1952)
- ☐ John (1952)
- ☒ McIntyre (1952)

Introduction:

Ranked set sampling (RSS) is an alternative to simple random sampling that can sometimes offer large improvements in precision. McIntyre (1952) introduced the basic concept of ranked set sampling in order to estimate the population means of pasture and forage yields. The RSS

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Horvitz and Thompson introduced an unbiased estimator in:

Select the correct option

Q. The Horvitz Thompson Estimator?

Answer:

➤ HT Estimator

- The general theory of unequal probabilities sampling without replacement was presented firstly by Horvitz and Thompson (1952).

 Reload Math Equations

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 12:55:50 PM, 21 July 2023)

Total Marks: 1

What is the R command to find out mean of x?

Select the correct option

 Reload Math Equations☒ mean(x)☐ mean*x

Show Answer

Click to Save Answer & Move to Next Question

Question # 2 of 10 (Start time: 01:06:13 PM, 14 February 2021)

Total Marks: 1

The Ratio Estimator for population total is described as:

Select the correct option

[Reload Math Equations](#)

- | | |
|----------------------------------|---|
| ☒ | $y_r = \frac{y}{x} X$ |
| ☐ | $y_r = \frac{y}{x}$ |
| ☐ | $\bar{y}_r = \frac{\bar{y}}{\bar{x}} \bar{X}$ |
| ☐ | $\bar{y}_r = \frac{\bar{y}}{\bar{x}} \bar{X}$ |

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Question # 3 of 10 (Start time: 01:07:59 PM, 14 February 2021)

Total Marks: 1

The

$$E(c_0^2)$$

is

Select the correct option

[Reload Math Equations](#)☐

$$\lambda C_x^2$$

☐

$$\lambda C_y^2$$

☐

$$\lambda C_y$$

☒

$$C_y x$$

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[Move to Next Question](#)

Question # 8 of 10 (Start time: 01:14:03 PM, 14 February 2021)

Total Marks: 1

$$E(e_0^2) = \text{-----}$$

where

Select the correct option

[Reload Math Equations](#)

- | | |
|----------------------------------|--|
| ☐ | λC_x^2 |
| ☐ | λC_y^2 |
| ☒ | $\left(\lambda C_y^2 + \frac{1}{\lambda} \right) C_x^2$ |
| ☐ | $\left(\lambda C_x^2 + \frac{1}{\lambda} \right) C_y^2$ |

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[Move to Next Question](#)



Question # 6 of 10 (Start time: 01:10:08 PM, 14 February 2021)

Total Marks: 1

What is the R command for writing bi-variate normal distribution?

Select the correct option

- ☐ `r=mvnorm(N,m,sig);`
- ☐ `r=mvnorm(N,m,sig);`
- ☐ `r=vnorm(N,m,sig);`
- ☒ `r=norm(N,m,sig);`

[Move to Next Question](#)

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Question # 9 of 10 (Start time: 01:14:19 PM, 14 February 2021)

Total Marks: 1

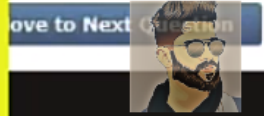
Zoomed to fill

The R command for defining a population is:

Select the correct option

- ☐ y<- c(2,8,7,9,12,11,18,16,12,19)
- ☐ y<- (2,8,7,9,12,11,18,16,12,19)
- ☒ y<- sample(2,8,7,9,12,11,18,16,12,19)
- ☐ y<- sub(2,8,7,9,12,11,18,16,12,19)

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Question # 10 of 10 (Start time: 01:14:55 PM, 14 February 2021)

Total Marks: 1

The ratio estimator for population mean is defined as:

Select the correct option

[Reload Math Equations](#)

- | | |
|----------------------------------|---|
| ☐ | $\bar{y}_1 = \frac{\bar{y}\bar{x}}{\bar{X}}$ |
| ☐ | $\bar{y}_2 = \frac{\bar{y}}{\bar{X}}$ |
| ☐ | $\bar{y}_3 = \frac{\bar{y}}{\bar{x}} \bar{X}$ |
| ☒ | None of the above |

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[Go to Next Question](#)



What is the value of combined type ratio estimator with

$$\bar{y}_{st}$$

=11.5.

Select the correct option

[Reload Math Equations](#)

- | | |
|----------------------------------|-------|
| ☒ | 6.988 |
| ☐ | 7.279 |
| ☐ | 8.301 |
| ☐ | 7.845 |

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What is the R command for writing variance-covariance matrix

$$\Sigma = \begin{bmatrix} 1 & 0.65 \\ 0.65 & 1 \end{bmatrix}$$

Select the correct option

[Reload Math Equations](#)

- | | |
|----------------------------------|---|
| ☒ | <code>sig=matrix(c(1,0.65,0.65,1),ncol=2)</code> |
| ☐ | <code>sig=matrix(c(1,0.65,0.65,1),ncol=4)</code> |
| ☐ | <code>sig=matrix(c(1,1,0.65,0.65),ncol=2);</code> |
| ☐ | None of the above |

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What is the mean of ratio estimator in two stage sampling?

Select the correct option

[Reload Math Equations](#)

- ☐ $MSE(\bar{y}_{r,2s}) \approx \bar{Y}^2 (V_{02} - V_{20} - 2V_{11})$
- ☐ $MSE(\bar{y}_{r,2s}) \approx \bar{Y}^2 (V_{02} + V_{20} - 2V_{11})$
- ☐ $MSE(\bar{y}_{r,2s}) \approx \bar{Y}^2 (V_{02} - V_{20} + 2V_{11})$
- ☐ None of the above

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What is the formula of ratio estimator?

Select the correct option

 Reload Math Equations

☒	$\frac{\bar{y}}{\bar{x}} \bar{X}$
☐	$\frac{\bar{x}}{\bar{y}} \bar{X}$
☐	$\frac{\bar{x}}{\bar{y}} \bar{Y}$
☐	None of the above

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Question # 5 of 10 (Start time: 12:03:13 PM, 24 February 2021)

Total Marks: 1

Which of the following is not an example of a nonrandom sampling technique?

Select the correct option

- ☒ Purposive
- ☐ Quota
- ☐ Cluster
- ☐ Convenience

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Go to Next Question



Question # 1 of 10 (Start time: 01:33:33 PM, 24 July 2023)

Total Marks: 1

For any complex number z , $\frac{d}{dz}(\cot z) = \underline{\hspace{2cm}}$.

Select the correct option

[Reload Math Equations](#)

- | | |
|----------------------------------|------------------|
| ☐ | $-\sec^2 z$ |
| ☒ | $-\csc^2 z$ |
| ☐ | $-\sec z \tan z$ |
| ☐ | $\tan z \sec z$ |

[Click to Save Answer & Move to Next Question](#)

Show Answer

Question # 2 of 10 (Start time: 01:35:23 PM, 24 July 2023)

Total Marks: 1

 $z^{\frac{1}{k}}$ is the _____ function.

Select the correct option

 Reload Math Equations☒ multivalued☐ single valued

Click to Save Answer & Move to Next Question

Show Answer

Question # 3 of 10 (Start time: 01:36:45 PM, 24 July 2023)

Total Marks: 1

For $z = x + i y$, we have $\sin z = \underline{\hspace{2cm}}$

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|-------------------------------------|
| ☐ | $\sin x \cosh y + i \cos x \sinh y$ |
| ☐ | $\sin x \cos y + i \cos x \sin y$ |
| ☒ | $\cos x \sin y + i \sin x \cos y$ |
| ☐ | $\cos x \sinh y + i \sin x \cosh y$ |

Click to Save Answer & Move to Next Question

Show Answer

Question # 4 of 10 (Start time: 01:38:27 PM, 24 July 2023)

Total Marks: 1

Range of principle argument of $\text{Log}(z)$ is _____

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---------------|
| ☐ | $(-\pi, \pi)$ |
| ☒ | $[-\pi, \pi]$ |
| ☐ | $(-\pi, \pi]$ |
| ☐ | $[-\pi, \pi)$ |



Click to Save Answer & Move to Next Question

SHOW ANSWER

Question # 5 of 10 (Start time: 01:39:56 PM, 24 July 2023)

Total Marks: 1

For $z = x + iy$, we have $\cos z =$ _____.

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|-------------------------------------|
| ☒ | $\cos x \cosh y - i \sin x \sinh y$ |
| ☐ | $\sin x \sinh y + i \cos x \cosh y$ |
| ☐ | $\sin x \cosh y + i \cos x \sinh y$ |
| ☐ | $\cos x \cosh y + i \sin x \sinh y$ |

Click to Save Answer & Move to Next Question

Show Answer

Question # 6 of 10 (Start time: 01:40:24 PM, 24 July 2023)

Total Marks: 1

The complex exponential function e^z is onto function.

Select the correct option

[Reload Math Equations](#)

True



False



Show Answer

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